

# Musthofa Joko Anggoro

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## RESEARCH INTERESTS

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My research interests lie in reconfigurable computing, computer architecture, and digital hardware design. I am particularly interested in the translation of architectural models into synthesisable hardware: spanning RTL design, hierarchical system decomposition, FPGA implementation, and hardware verification. More broadly, I am drawn to research that bridges theoretical computing models with practical, resource-constrained hardware realisation on programmable logic platforms.

## EDUCATION

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**University of Indonesia (UI) — Faculty of Computer Science**

Jakarta, Indonesia

*Bachelor of Computer Science*

*Aug 2022 – Aug 2026 (Expected)*

- **GPA: 3.84 / 4.00**
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

**Nanyang Technological University (NTU) & ITS**

Singapore & Surabaya

*INSPIRASI-NTU Summer Programme 2025 — LPDP Scholarship Awardee*

*July 2025*

- Applied software engineering and systems principles to sustainable development; produced a technical report proposing a tiered smart-port automation framework with IoT sensor networks and renewable energy integration

## UNDERGRADUATE THESIS

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**VHDL-93 Implementation of the Mano Register-Transfer Microoperation Model on FPGA**

*Supervisor:* Prof. Petrus Mursanto, University of Indonesia

- Designed and implemented the Mano register-transfer computer model as a complete, hierarchically structured VHDL-93 datapath on a **Xilinx Spartan-3AN** FPGA
- Conducted a systematic **gap analysis** mapping undefined textbook constraints to concrete RTL design decisions, bridging theory and physical hardware implementation
- Verified all modules via **ISim functional simulation** testbenches and achieved **timing closure at 50 MHz** (critical path: 17.05 ns) using XST synthesis
- *Toolchain:* Xilinx ISE 14.x · XST · ISim · JTAG · VHDL-93

## PUBLICATIONS

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- Musthofa Joko Anggoro, Petrus Mursanto. “VHDL-93 Implementation of the Mano Register-Transfer Microoperation Model on an FPGA.” *Proceedings of the International Conference on Advanced Computer Science and Information Systems (ICACSIS) 2026*. [In preparation]